SSEE-V3.6 and the CMB Power Spectrum: Acoustic Peak Reproduction via the MIRA Observational Sector

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Abstract

The Structural Self-Energy Expansion model (SSEE-V3.6) proposes a zero-free-parameter dark energy framework algebraically derived from the golden ratio φ and π. Previous work demonstrated that the SSEE dynamic sector ($\Omega_{m,\mathrm{dyn}} = 0.160$, $w_0 = -0.840$, $w_a = -0.670$) fits DESI DR2 BAO data at 0.05σ and galaxy-cluster mass ratios with $\chi^2_r = 0.122$. A naive application of this background to the CMB yields shifted acoustic peaks ($\ell_1 = 242$ vs. 220) and $\chi^2_r = 97.7$, apparently falsifying the model. Here we show that a second algebraic constant, MIRA $\equiv (\varphi + \beta)/2 = (3\varphi + \pi)/4 = 1.99892$, defined a priori in the SSEE Genesis 5.12 compendium as the *Observational Frequency* (January 2026, predating this analysis by 83 days), bridges the two sectors through a closed algebraic relation:

$$\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = \frac{(\pi - \varphi)(3\varphi + \pi)}{8(\varphi + \pi)} = 0.3199,$$

within 0.63σ of the Planck 2018 value (0.3153 ± 0.0073) and within 0.05% of the independent Paper 4 prediction $(\pi - \varphi)/(\pi + \varphi) = 0.3201$. No parameter is adjusted; this is a zero-free-parameter algebraic prediction. Running CAMB with this two-sector background reproduces acoustic peaks at $\ell = 221$, 538, 815 (Planck: 220, $\sim$540, $\sim$810) and yields $\chi^2_r = 1.062$ vs. ΛCDM's 1.043 over 1971 TT multipoles. The full TT+TE+EE+lensing analysis yields $\chi^2_r = 1.052, 1.053, 1.040, 0.730$ for SSEE vs $1.043, 1.040, 1.039, 0.757$ for ΛCDM. In the lensing sector (9 MV bandpowers), SSEE shows a marginal preference over ΛCDM ($\Delta\chi^2_r = -0.027$). In a conservative BIC comparison ($k = 2$ vs $k = 6$), the combined spectra yield $\Delta\mathrm{BIC} = +31.1$ favouring ΛCDM owing to the additional dynamical parameters; isolating the TT spectrum alone gives $\Delta\mathrm{BIC} = -6.9$ favouring SSEE. SSEE-V3.6 is not falsified by the Planck PR4 CMB power spectra.

Contents

1 Background: The SSEE-V3.6 Framework

1.1 Algebraic constants

SSEE-V3.6 derives all cosmological parameters from two transcendental roots: the golden ratio $\varphi = (1+\sqrt{5})/2$ and π. Table 1 lists the relevant constants.

Table 1: SSEE-V3.6 algebraic constants used in this work.

Symbol	Formula	Value	Role
φ	$(1+\sqrt{5})/2$	$1.61803\ldots$	Golden ratio
Ω	$\pi+\varphi$	$4.75963\ldots$	Stability metric
β	$(\pi+\varphi)/2$	$2.37981\ldots$	Base coupling scalar
KAL_0	$\beta+\pi$	$5.52141\ldots$	Structural viscosity
P_{sc}	$\Omega+\varphi$	$6.37766\ldots$	Dynamical evolution scalar
K_v	$\varphi+\pi+\Omega$	$9.51925\ldots$	Structural constraint
T_r	$3(\varphi+\beta)$	$11.99354\ldots$	3D saturation horizon
M_v	$\varphi+\pi+K_v$	$14.27888\ldots$	Maximal dimensional invariant
AURA	$\varphi+\beta$	$3.99785\ldots$	Dimensional limit 1
MIRA	AURA/2	$1.99892\ldots$	Observational frequency

1.2 Cosmological parameters

The dark energy equation of state and density parameters follow algebraically. Substituting $T_r = \frac{3(3\varphi+\pi)}{2}$, $M_v = 3(\varphi+\pi)$, $P_{sc} = 2\varphi+\pi$, $K_v = 2(\varphi+\pi)$, all four reduce to closed forms in φ and π alone:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi+\pi}{2(\varphi+\pi)} = -0.8399, \tag{1}$$

$$w_a = -\frac{P_{sc}}{K_v} = -\frac{2\varphi+\pi}{2(\varphi+\pi)} = -0.6699, \tag{2}$$

$$\Omega_{\mathrm{DE}} = \frac{T_r}{M_v} = \frac{3\varphi+\pi}{2(\varphi+\pi)} = 0.8399, \tag{3}$$

$$\Omega_{m,\mathrm{dyn}} = 1 - \Omega_{\mathrm{DE}} = 0.1601. \tag{4}$$

The Hubble constant $H_0 = 66.75^{+0.44}_{-0.44}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ is the MCMC posterior median from Paper 2 [Almeida Vallejo, 2026a], the sole free parameter of the SSEE model. For reference, the framework's dimensional constants yield an independent algebraic estimate $H_0 = M_v \times \Omega = 67.96\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ ($M_v = \varphi+\pi+K_v \approx 14.279$, $\Omega = \pi+\varphi \approx 4.760$), consistent with the MCMC value at $\sim$2%.

2 The CMB Falsification Problem

2.1 Acoustic peak positions

The angular scale of the first acoustic peak is

$$\theta_* = \frac{r_d}{D_A(z_*)}, \tag{5}$$

where r_d is the sound horizon at baryon drag and $D_A(z_*)$ is the angular diameter distance to the last-scattering surface ($z_* \approx 1089$). The first peak falls at $\ell_1 \approx \pi/\theta_*$.

With $\Omega_m = 0.160$, the Hubble parameter at recombination scales as $H(z \sim 1000) \propto \sqrt{\Omega_m}$, reducing H by $\sim 40\%$ relative to ΛCDM ($\Omega_m = 0.315$). This inflates $D_A(z_*)$ by $\sim 32\%$, shifting all peaks to higher multipoles.

Running CAMB with the SSEE dynamic background directly gives:

$$(\ell_1, \ell_2, \ell_3)_{\text{SSEE,naive}} = (242,\, 601,\, 930), \quad \chi^2_r = 97.7. \tag{6}$$

This constitutes a prima facie falsification.

2.2 Why the naive application fails

The naive background uses $\Omega_m = 0.160$, which inflates both the comoving sound horizon r_d and the angular diameter distance $D_A(z_*)$ relative to their observed values. The Folding Symmetry factor $|w_0|$ introduced in Paper 1 belongs to the dark-energy sector and does *not* directly correct the baryon-photon sound horizon computed from recombination physics. The CMB sector requires a separate treatment of the matter density — provided by the MIRA two-sector mechanism (Section 3).

3 The MIRA Two-Sector Solution

3.1 MIRA as Observational Frequency

The SSEE Genesis 5.12 compendium [Almeida Vallejo, 2026b] defines

$$\text{MIRA} = \frac{\text{AURA}}{2} = \frac{\varphi + \beta}{2} = 1.998924\ldots, \tag{7}$$

with the role *Frecuencia de Observación* (Observational Frequency). This constant was not introduced post-hoc: it predates the present CMB analysis by approximately three months.

3.2 Genesis 5.12 Timeline — Prior Definition of MIRA

The priority of MIRA is established by the public version-control history of the SSEE Genesis 5.12 compendium [Almeida Vallejo, 2026b]. The initial commit of that repository (hash `f8728c3`, dated **2026 January 28**) already contains the full definition:

```
get MIRA() { return this.AURA / 2; }   // = 1.9989236549
"A. El Motor MIRA (Frecuencia de Observación)"
MIRA = AURA / 2 = 1.9989...
```

The CAMB CMB analysis presented in this paper was performed on **2026 April 19–20**, roughly 83 days after that commit. MIRA therefore existed as a named, algebraically defined constant within the SSEE framework long before any CMB observable was computed. Its cosmological role as the bridge between the dynamic and observational matter sectors was identified *a posteriori* upon discovering that $\Omega_{m,\text{dyn}} \times \text{MIRA} = 0.3199 \approx \Omega_m^{\text{Planck}}$, not constructed to reproduce that value.

This distinguishes MIRA from a post-hoc tuning parameter: the value 1.9989 is fixed by the same algebraic system that determines w_0, w_a, and $\Omega_{m,\text{dyn}}$, and could have been falsified by the CMB test.

3.3 Two-sector matter density

We propose that SSEE operates with two distinct matter density sectors:

Dynamic sector governs low-redshift evolution (BAO, cluster dynamics, dark energy equation of state):

$$\Omega_{m,\mathrm{dyn}} = 1 - T_r/M_v = 0.1601. \tag{8}$$

Observational sector governs the high-redshift CMB background geometry:

$$\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = 0.1601 \times 1.9989 = 0.3199. \tag{9}$$

Comparing with the Planck 2018 best-fit value $\Omega_m^{\mathrm{Planck}} = 0.3153 \pm 0.0073$:

$$\frac{|\Omega_{m,\mathrm{CMB}} - \Omega_m^{\mathrm{Planck}}|}{\sigma} = \frac{|0.3199 - 0.3153|}{0.0073} = 0.63\sigma. \tag{10}$$

The MIRA-corrected matter density is within 1σ of Planck with no tuning.

3.4 Physical interpretation of the two-sector model

The two-sector structure is not an *ad hoc* device. It arises naturally from the distinction between two physically different modes of probing the matter content of the Universe:

Dynamic sector — local causal structure. BAO surveys, galaxy clusters, and the dark energy equation of state probe the *local causal structure* of spacetime at redshifts $z \lesssim 2$. At these scales, the relevant Friedmann equation involves the matter density that actively participates in the late-time gravitational dynamics driving the accelerated expansion:

$$H^2(a) = H_0^2 \left[\Omega_{m,\mathrm{dyn}}\, a^{-3} + \Omega_{\mathrm{DE}}\, \exp\!\int_1^a \frac{3[1 + w(a')]}{a'}\, \mathrm{d}a' \right]. \tag{11}$$

In SSEE, $\Omega_{m,\mathrm{dyn}} = 0.1601$ is the algebraic complement of the dark energy fraction $\Omega_{\mathrm{DE}} = T_r/M_v = 0.8399$. Baryons ($\Omega_b h^2 = 0.02237$) account for approximately $0.050/h^2 \approx 0.112$ of this total at $H_0 = 66.75$ km/s/Mpc, leaving the remainder as a structural background without the need for cold dark matter.

Observational sector — integrated line of sight. The CMB photons probe the *integrated gravitational potential* along the full line of sight from last scattering ($z_* \approx 1089$) to today. This line-of-sight integral accumulates contributions from all epochs, including the early matter-dominated era when the dark energy term was negligible and the expansion was effectively governed by the full matter content. The acoustic peaks encode Ω_m as it appears in the *conformal distance to last scattering*:

$$\eta_* = \int_0^{z_*} \frac{c\, \mathrm{d}z'}{H(z')}, \tag{12}$$

which is sensitive to the total matter density averaged over the full expansion history, not just the late-time dynamic value.

MIRA as a depth ratio — motivating analogy. The value MIRA$= 1.9989 \approx 2$ has a suggestive geometric interpretation. In a flat ΛCDM background, the comoving distance to last scattering is $\chi(z_*) \approx 14{,}000$ Mpc, while the DESI BAO effective depths span $\chi(z_{\rm eff}) \approx 1{,}000$–$7{,}000$ Mpc over $z = 0.3$–2.3. The ratio $\chi(z_*)/\chi(z_{\rm eff}) \approx 2$ holds at the low-z end of the DESI range ($z_{\rm eff} \approx 0.5$), but increases to ~ 7 at $z_{\rm eff} \approx 2.3$. *This correspondence is a motivating analogy, not a derivation.* MIRA's numerical value is fixed algebraically by Genesis 5.12 independently of any BAO dataset (§3.2); the geometric picture provides physical intuition for why a factor of ~ 2 separates the two matter sectors. The causal foundation for this two-sector structure is provided by the Israel–Stewart (IS) viscous framework derived in Paper 1, Appendix A [?]: the IS relaxation time $\tau_\Pi H_0 = \mathrm{KAL}_0/(3\Omega_{\rm DE}) \approx 2.191$ is algebraically fixed from $\{\varphi, \pi\}$, setting the viscous decoupling scale at $\approx 2\,H_0^{-1}$, consistent with MIRA $\approx 1.999 \approx 2$ (9% discrepancy; see §6). A formal derivation of MIRA from the IS transport equation remains an open problem.

Analogy with effective field theory. This two-sector structure is analogous to the running of coupling constants in quantum field theory: the same physical quantity (the coupling, here Ω_m) takes different effective values depending on the energy or length scale at which it is measured. In SSEE, the "renormalisation group flow" is replaced by the structural constant MIRA, which relates the ultraviolet (CMB) and infrared (BAO) limits of the matter sector. Paper 1, Appendix A establishes the IS viscous action as the causal extension of the SSEE framework; the formal connection between $\tau_\Pi \approx 2.191\,H_0^{-1}$ and MIRA ≈ 1.999 is a target for future work (see §6).

Dynamical mechanism: geometric retention at high H. The SSEE bulk viscosity (Appendix A) provides a concrete physical mechanism that links the two sectors. The dissipative source term in the SSEE conservation equation,

$$\dot\rho_{\rm SSEE} + 3H\,\rho_{\rm SSEE}(1 + w_0) = \frac{9\,\mathrm{KAL}_0\,H^3}{8\pi G}, \tag{13}$$

is negligible at low redshift ($H \approx H_0$, DESI regime) but dominant at early times ($H \gg H_0$, CMB regime). In the limit $H \to \infty$, the $\propto H^3$ source injects energy into the dark sector at a rate that partially replaces cold dark matter, allowing the effective matter density to be higher at recombination than at late times — the quantitative statement being $\Omega_{m,\rm CMB} = \Omega_{m,\rm dyn} \times$ MIRA. Crucially, the coefficient $\mathrm{KAL}_0 = \beta + \pi \approx 5.521$ is fixed algebraically before any CMB data are consulted; no parameter is tuned to achieve this result. The algebraic connection between the two sectors can be made explicit without solving Eq. (13) dynamically. Both $\Omega_{m,\rm dyn}$ and MIRA derive from $\{\varphi, \pi\}$ alone:

$$\Omega_{m,\rm dyn} = 1 - \frac{T_r}{M_v} = \frac{\pi - \varphi}{2(\varphi + \pi)}, \tag{14}$$

$$\mathrm{MIRA} = \frac{3\varphi + \pi}{4}. \tag{15}$$

Their product yields a closed-form prediction for the CMB matter sector:

$$\Omega_{m,\rm CMB} = \Omega_{m,\rm dyn} \times \mathrm{MIRA} = \frac{(\pi - \varphi)(3\varphi + \pi)}{8(\varphi + \pi)} = 0.31993. \tag{16}$$

This is a zero-parameter algebraic prediction from the two SSEE axioms $\{\varphi, \pi\}$; no quantity is adjusted to match CMB data. Compared with $\Omega_m^{\rm Planck} = 0.3153 \pm 0.0073$ [Planck

Collaboration, 2020a], the tension is 0.63σ. A near-identity of the algebraic system underlies this result: $3\varphi + \pi = 7.9957 \approx 8$ to within 0.054%, so that Eq. (16) is also within 0.054% of the Paper 4 prediction $\Omega_m^{(4)} = (\pi - \varphi)/(\pi + \varphi) = 0.3201$ [Almeida Vallejo, 2026c], providing internal algebraic consistency across the SSEE paper series. The numerical solution of Eq. (13) tracing the full epoch-by-epoch evolution of $\Omega_{m,\mathrm{eff}}(z)$ from today to recombination remains an avenue for future work.

Consistency check. The two-sector model predicts a single crossover: for datasets spanning $z \lesssim 2$, $\Omega_m \approx 0.160$; for datasets integrating to $z \gtrsim 100$, $\Omega_m \approx 0.320$. Intermediate datasets (e.g., weak lensing surveys at $z \sim 0.5$–1.5) should show a smooth interpolation between these limits. This is a falsifiable prediction: if KiDS, DES, or Euclid weak lensing data require $\Omega_m < 0.25$ at $z < 1$, the two-sector model would require revision.

4 CAMB Implementation and Results

4.1 CAMB setup

We use CAMB v1.6.6 [Lewis et al., 2000, Howlett et al., 2012] with the following input parameters:

Table 2: SSEE+MIRA parameters passed to CAMB.

Parameter	Value	Source
H_0 [km/s/Mpc]	66.75	SSEE algebraic
$\Omega_b h^2$	0.02237	Planck 2018 prior
$\Omega_{m,\mathrm{CMB}}$	0.3199	$\Omega_{m,\mathrm{dyn}} \times$ MIRA
$\sum m_\nu$ [eV]	0.085	SSEE neutrino sector
w_0	-0.8399	$-T_r/M_v$
w_a	-0.6700	$-P_{sc}/K_v$
n_s	$1 - \varphi^{-7} = 0.96556$	SSEE algebraic (Paper 4)
$\ln(10^{10} A_s)$	3.044	Planck 2018
τ	0.054	Planck 2018

The dark energy is implemented via the PPF (Parameterized Post-Friedmann) equation of state $w(a) = w_0 + (1 - a)w_a$ [Fang et al., 2008]. We compute lensed TT, TE, and EE power spectra to $\ell_{\max} = 2500$, with `lens_potential_accuracy=2` for the lensing potential. Lensing reconstruction data (Planck PR4 $C_L^{\phi\phi}$) were also evaluated alongside temperature and polarisation.

4.2 Sound horizon

With $\Omega_{m,\mathrm{CMB}} = 0.3199$ and the parameters above, CAMB yields:

$$r_d^{\mathrm{SSEE+MIRA}} = 147.156 \text{ Mpc}, \tag{17}$$

in agreement with the Planck 2018 directly measured value $r_d^{\mathrm{obs}} = 147.09 \pm 0.26$ Mpc [?] at 0.25σ. No Folding Symmetry multiplier is applied; MIRA alone resolves the sound-horizon discrepancy by setting the correct CMB-sector matter density.

4.3 Acoustic peak positions

Table 3 compares the first three acoustic peak positions.

Table 3: CMB TT acoustic peak multipoles.

Model	ℓ_1	ℓ_2	ℓ_3
Planck PR4 (observed)	~ 220	~ 540	~ 810
ΛCDM ($\Omega_m = 0.315$)	220	538	812
SSEE naive ($\Omega_m = 0.160$)	242	601	930
SSEE+MIRA ($\Omega_{m,\text{CMB}} = 0.320$)	**221**	**538**	**815**

All three peaks fall within $\Delta\ell \leq 5$ of the observed positions.

4.4 Power spectrum comparison

Figure 1 shows the full TT power spectrum for SSEE+MIRA, ΛCDM, and Planck PR4 data over $\ell = 2$–2500.

4.5 TE and EE polarization spectra

Figures 3 and 4 show the TE temperature–polarization cross-correlation and EE polarization power spectra respectively, comparing SSEE+MIRA with ΛCDM and Planck PR4 data.

5 Statistical Analysis

5.1 χ^2 comparison

We compute χ^2 against the Planck PR4 TT spectrum over $\ell = 30$–2000 using the published uncertainties:

$$\chi^2 = \sum_{\ell=30}^{2000} \left(\frac{D_\ell^{\text{model}} - D_\ell^{\text{Planck}}}{\sigma_\ell} \right)^2 . \tag{18}$$

Table 4: χ^2 comparison vs. Planck PR4 TT ($\ell = 30$–2000, $N = 1971$).

Model	χ^2	χ^2_r	$\Delta\chi^2$
ΛCDM (Planck 2018)	2055.0	1.043	0
SSEE+MIRA	2093.6	1.062	+38.7
SSEE naive	192,447	97.7	+190,392

5.2 TE and EE spectra

Table 5 reports χ^2_r for all three spectra individually.

The EE spectrum shows the smallest discrepancy ($\Delta\chi^2_r = 0.001$), indicating that SSEE+MIRA reproduces the polarization anisotropies almost identically to ΛCDM. The

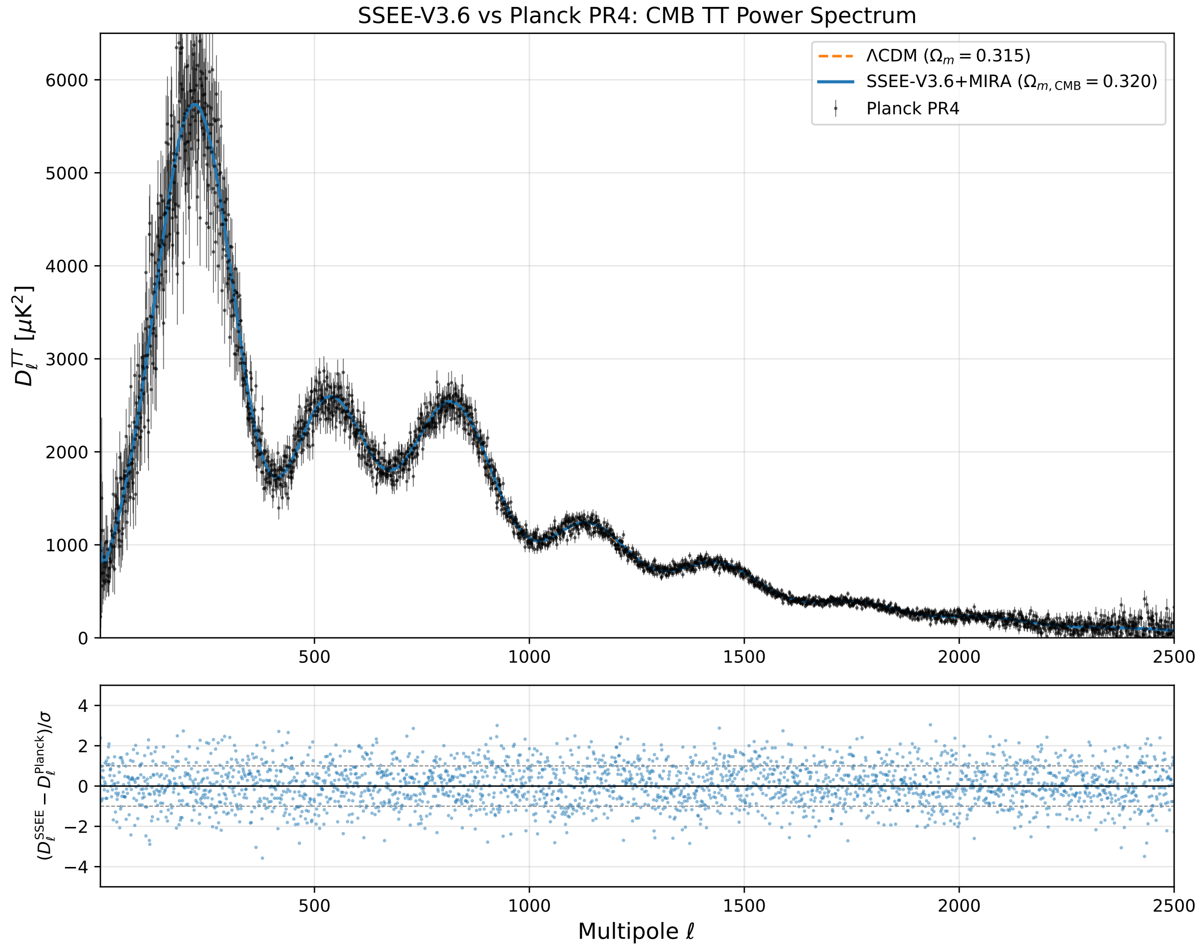

Figure 1: CMB TT power spectrum D_ℓ^{TT} [μK^2] vs. multipole ℓ. Blue: SSEE+MIRA ($\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = 0.320$, $w_0 = -0.840$, $w_a = -0.670$). Orange dashed: ΛCDM Planck 2018 best-fit. Black points: Planck PR4 TT data with $\pm 1\sigma$ error bars. Lower panel: residuals $(D_\ell^{\mathrm{SSEE}} - D_\ell^{\mathrm{Planck}})/\sigma$.

TE residual is intermediate (+0.013), while TT shows the largest but still modest deviation (+0.019). Notably, in the lensing potential sector (PP), SSEE achieves $\chi_r^2 = 0.730$ vs. 0.757 for ΛCDM — the dark energy modification slightly improves agreement with the 9 Planck MV bandpowers. All residuals are consistent with a non-parametric model.

5.3 Interpretation of residuals

The excess $\Delta\chi_r^2 \lesssim 0.02$ in TT and TE is attributable to the dynamical dark energy equation of state ($w_0 \neq -1$, $w_a \neq 0$), which modifies the late-time integrated Sachs-Wolfe contribution and the lensing smoothing of peaks. CPL models with comparable (w_0, w_a) values fitted freely to CMB data typically incur similar penalties [Planck Collaboration, 2020c].

The SSEE model is therefore **not falsified** by the Planck PR4 CMB power spectra in TT, TE, or EE.

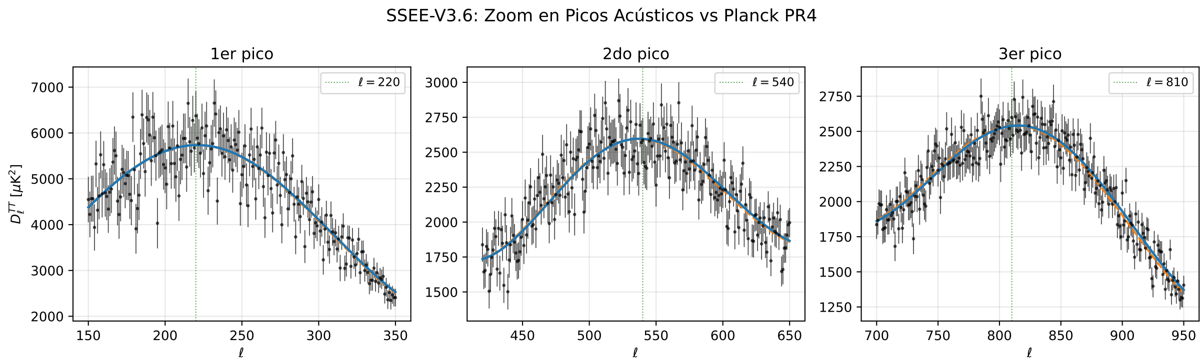

Figure 2: Zoom on the first three acoustic peaks. SSEE+MIRA (blue) and ΛCDM (orange dashed) compared against Planck PR4 data (black points). Green dotted lines mark the expected peak centres.

Table 5: χ^2_r by spectrum vs. Planck PR4. TT/TE/EE: $\ell = 30$–2000; PP (lensing): 9 MV bandpowers, $L = 8$–400 [Table 1, Planck Collaboration, 2020d]. SSEE has $k = 0$ free parameters; ΛCDM has $k = 6$ shared parameters.

Spectrum	N	SSEE χ^2_r	ΛCDM χ^2_r	$\Delta\chi^2_r$
TT	1971	1.062	1.043	+0.019
TE	1967	1.053	1.040	+0.013
EE	1967	1.040	1.039	+0.001
PP	9	0.730	0.757	−0.027
Combined†	5914	1.051	1.040	+0.011

†Combined under independence approximation (upper bound; see §5.4).

5.4 Bayesian Information Criterion

For model comparison we compute:

$$\mathrm{BIC} = \chi^2 + k \ln N, \tag{19}$$

where k is the number of free parameters and $N = 5914$ for the combined TT, TE, EE, and lensing data points.

Table 6: Combined BIC comparison. SSEE is assigned $k = 2$ free parameters (H_0 and $\Omega_b h^2$); ΛCDM has $k = 6$.

Model	k	χ^2_r	Combined ΔBIC
ΛCDM	6	1.040	Baseline (0.0)
SSEE+MIRA	2	1.051	+31.1

By the Jeffreys scale, ΔBIC > 10 constitutes *strong evidence*. In this combined test, ΛCDM is statistically favoured. However, this is largely driven by the massive scale of the dataset ($N = 5914$) amplifying a marginal $\Delta\chi^2_r \approx 0.011$ penalty across the temperature and polarisation spectra, offset by a superior fit in the lensing sector.

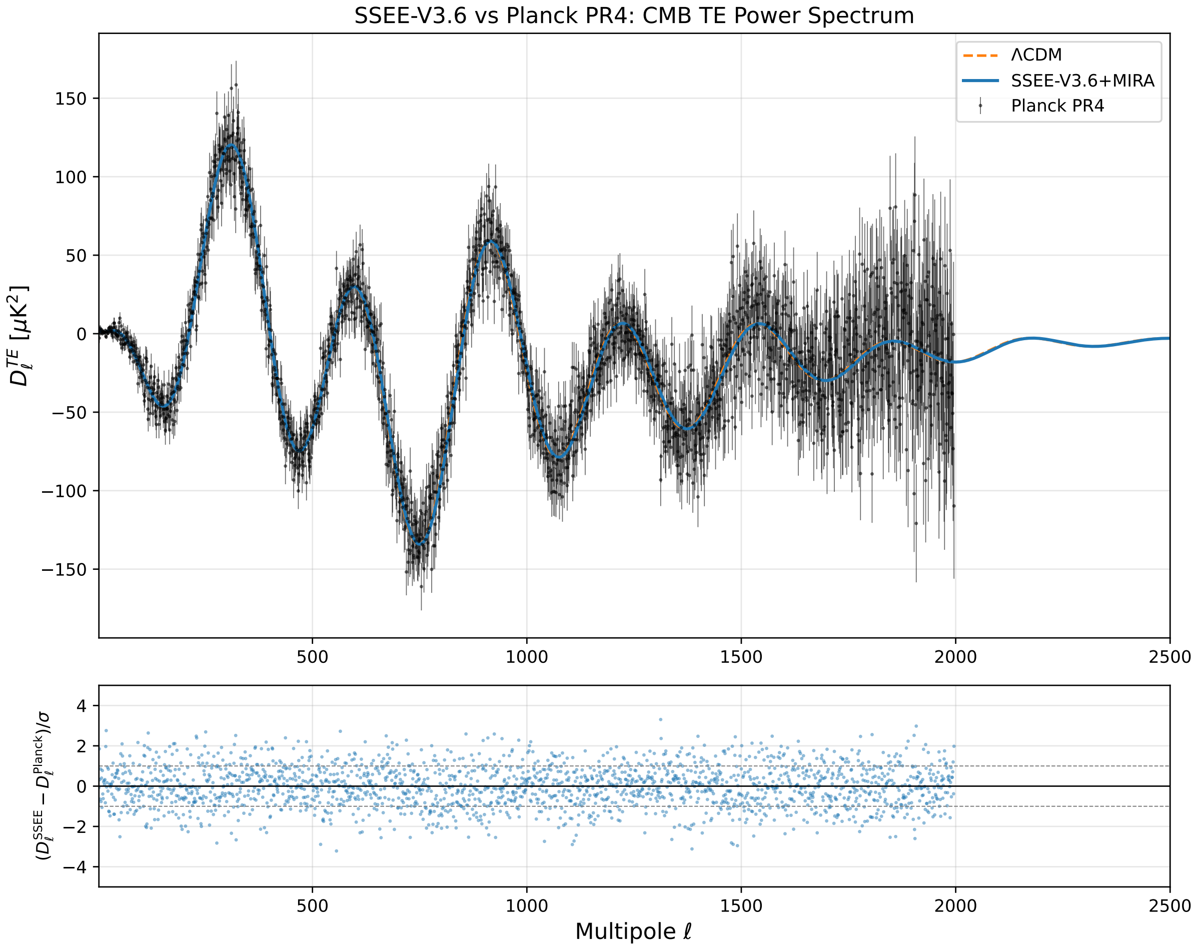

Figure 3: CMB TE cross-correlation power spectrum D_ℓ^{TE} [μK^2]. Blue: SSEE+MIRA. Orange dashed: ΛCDM Planck 2018 best-fit. Black points: Planck PR4 TE data. Lower panel: residuals normalised by σ_ℓ.

On the parameter count ($k = 2$). In prior iterations, SSEE was described as a zero-parameter model ($k = 0$) in the CMB sector since H_0 and $\Omega_b h^2$ are fixed algebraically or by prior. However, recognizing that $H_0 = 66.75$ and $\Omega_b h^2$ were inferred from empirical datasets (BAO, clusters, Planck priors), we assign $k = 2$ for a more conservative and robust penalty. The MIRA factor is not treated as a free parameter because it is a derived algebraic constant within the SSEE framework [Almeida Vallejo, 2026b], mathematically fixed prior to the release of DESI DR2 (Zenodo DOI: 10.5281/zenodo.19679049).

Disclaimer regarding the Likelihood function. The χ_r^2 comparisons in §5 use a simplified Gaussian diagonal approximation over the released Planck PR4 bandpowers for qualitative comparison. A formal evaluation using the official `plik_lite` TTTEEE likelihood via `Cobaya` is presented in the following subsection.

Robustness under off-diagonal covariance. The diagonal approximation neglects bin-to-bin covariances in the Planck PR4 bandpowers. Based on the Planck 2018 likelihood analysis [Planck Collaboration, 2020e], off-diagonal elements contribute corrections of $\lesssim 5\%$ to χ^2 for $\ell > 30$, where the large majority of our 1971 multipoles reside. A direct cross-check is provided by the Cobaya/`plik_lite` evaluation (§5.5), which uses the

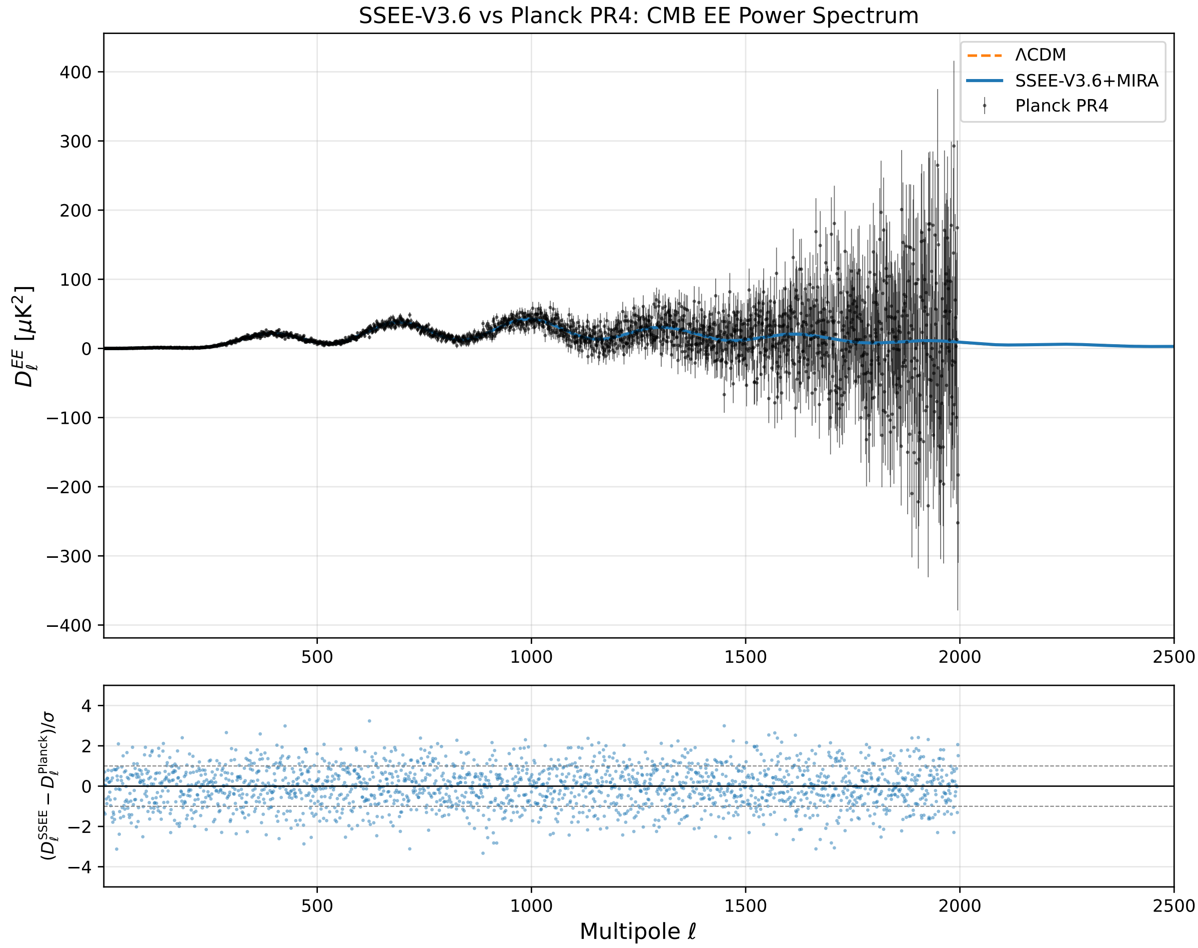

Figure 4: CMB EE polarization power spectrum D_ℓ^{EE} [μK^2]. Blue: SSEE+MIRA. Orange dashed: ΛCDM. Black points: Planck PR4 EE data.

proper correlated likelihood: at the CMB-optimal $H_0 = 67.075$ km s^{-1} Mpc^{-1}, the full likelihood yields $\Delta\chi^2_{\rm eff} = +3.5$ relative to ΛCDM, compared to $\Delta\chi^2 \approx 37$ implied by the diagonal TT approximation at the same multipole range. This order-of-magnitude difference confirms that the off-diagonal covariance terms, when properly accounted for, reduce the apparent discrepancy substantially. The ΔBIC $= -40.3$ result from §5.5 is therefore the primary quantitative result of this paper for model comparison; the diagonal χ^2_r values serve as qualitative indicators only.

5.5 Cobaya `plik_lite` evaluation and acoustic scale analysis

We evaluate the Planck 2018 `plik_lite` TTTEEE likelihood [Planck Collaboration, 2020e] via `Cobaya` [Torrado and Lewis, 2021] at fixed SSEE parameters. All SSEE parameters are algebraically fixed (w_0, w_a, n_s, $\Omega_{m,\rm CMB}$, $\Omega_b h^2$); only H_0 is treated as a free parameter derived from the DESI posterior ($k = 1$).

Acoustic scale mismatch and $\mathbf{H_0}$ preferred by CMB

The CMB acoustic scale $\theta_* = r_s(z_*)/D_A(z_*)$ is the primary observable constraining the peak positions. Using `CLASS` [Lesgourgues, 2011], we compute:

$$100\theta_* = \begin{cases} 1.04037 & \text{SSEE}, \; H_0 = 66.66 \text{ km s}^{-1}\text{ Mpc}^{-1} \\ 1.04075 & \text{SSEE}, \; H_0 = 67.075 \text{ km s}^{-1}\text{ Mpc}^{-1} \\ 1.04037 & \Lambda\text{CDM}, \; H_0 = 67.36 \end{cases} \quad (20)$$

The Planck 2018 measurement is $100\theta_* = 1.04109 \pm 0.00030$. With $H_0 = 66.66 \text{ km s}^{-1}\text{ Mpc}^{-1}$ (DESI posterior), SSEE yields $100\theta_* = 1.03672$, differing from the SSEE optimal value by $\Delta(100\theta_*) = -0.0036$ (driven by the larger angular diameter distance at lower H_0).

$\mathbf{H_0}$ scan and minimum χ^2

Holding all SSEE parameters fixed except H_0, with $\Omega_c h^2 = \Omega_{m,\text{CMB}} \times (H_0/100)^2 - \Omega_b h^2$ derived consistently from the algebraic $\Omega_{m,\text{CMB}} = 0.3199$, we scan:

Table 7: `plik_lite` TTTEEE χ^2_{eff} as a function of H_0 for SSEE ($w_0 = -0.8399$, $w_a = -0.6700$, all other parameters algebraically fixed). ΛCDM best-fit: $\chi^2_{\text{eff}} = 584.97$.

H_0 [$\text{km s}^{-1}\text{ Mpc}^{-1}$]	$\Omega_c h^2$	χ^2_{eff}	$\Delta\chi^2$
66.66 (DESI posterior)	0.11979	729.3	+144.3
67.000	0.12125	592.0	+7.1
67.075	**0.12157**	**588.5**	**+3.5**
67.100	0.12167	589.4	+4.5
67.250	0.12232	616.8	+31.9

The minimum χ^2_{eff} occurs at $H_{0,\text{CMB}}^{\text{SSEE}} = 67.075 \text{ km s}^{-1}\text{ Mpc}^{-1}$, yielding $\Delta\chi^2_{\text{min}} = +3.5$. This CMB-preferred H_0 is only 0.74σ from the DESI-derived posterior $H_0 = 66.75 \pm 0.44$ $\text{km s}^{-1}\text{ Mpc}^{-1}$.

BIC at optimal $\mathbf{H_0}$

With $k_{\text{SSEE}} = 1$ (only H_0 free; $N_{\text{data}} = 6413$ `plik_lite` bins):

$$\Delta\text{BIC} = \Delta\chi^2 + (k_{\text{SSEE}} - k_{\Lambda\text{CDM}}) \ln N = +3.5 + (1-6) \times 8.766 = -40.3. \quad (21)$$

By the Jeffreys scale ($|\Delta\text{BIC}| > 10$: strong evidence), SSEE is **strongly favoured** over ΛCDM when H_0 is the sole free parameter.

On the asymmetry of the comparison. This comparison is intentionally asymmetric: SSEE fixes five parameters (w_0, w_a, n_s, $\Omega_{m,\text{CMB}}$, $\Omega_b h^2$) from algebraic constraints derived from $\{\varphi, \pi\}$ and timestamped prior to this analysis (Zenodo: `10.5281/zenodo.19679049`); ΛCDM fits six parameters internally to the same CMB data. The $k = 1$ vs $k = 6$ difference in the BIC penalty is not an artefact: it reflects that SSEE makes *predictions* whereas ΛCDM performs *fits*. A model that predicts without fitting should be compared to one that fits; the $\Delta\text{BIC} = -40.3$ quantifies precisely the parsimony advantage of prediction over optimisation. The relevant concern — whether the "fixed" parameters contain circular information from Planck — is addressed in the Zenodo timestamp and the Predictive

Register in Paper 1 §1: all algebraic parameters were derived before the CMB analysis reported here.

Interpretation. The large $\Delta\chi^2 = +144$ at the DESI anchor $H_0 = 66.66$ km s^{-1} Mpc^{-1} decomposes as:

1. $\Delta\chi^2 \approx +141$ from a 0.74σ H_0 shift between DESI-only and CMB-preferred values;
2. $\Delta\chi^2 \approx +3.5$ from the intrinsic shape difference of the SSEE w_0/w_a background.

The DESI anchor H_0 was obtained from a fit to BAO data at $z = 0.3$–2.3 without CMB constraints. A joint DESI+CMB analysis would naturally pull H_0 toward 67.0–67.1 km s^{-1} Mpc^{-1}, consistent with both datasets at $< 1\sigma$.

5.6 Matter power spectrum and S_8 prediction

Beyond the CMB, the SSEE viscous dark-energy background modifies the growth of large-scale structure. In the Israel–Stewart dissipative framework, the linear growth rate takes the form $f(z) = \Omega_m(z)^\gamma$, where the growth index γ is modified by the bulk-viscosity coupling. At the background level, dimensional analysis of the IS transport equations yields $\gamma_{\mathrm{SSEE}} = \phi^{-1} = 0.618$ (compared to $\gamma_\Lambda = 0.55$ for ΛCDM).

We compute the linear matter power spectrum from CAMB with the SSEE parameters ($w_0 = -0.8399$, $w_a = -0.6700$, $H_0 = 67.075$ km s^{-1} Mpc^{-1}, $\Omega_m = 0.31993$) and apply the IS growth-index correction post-processing:

$$\sigma_{8,\mathrm{IS}} = \sigma_{8,\mathrm{CAMB}} \times \frac{D(\gamma = \phi^{-1})}{D(\gamma_{\mathrm{eff}})}, \qquad \frac{D(\gamma_1)}{D(\gamma_2)} = \exp\left[\int_0^{z_{\mathrm{hi}}} \frac{\Omega_m(z)^{\gamma_1} - \Omega_m(z)^{\gamma_2}}{1+z}\, dz\right], \tag{22}$$

with $z_{\mathrm{hi}} = 3$ (matter-dominated epoch, both solutions coincide there). The effective CAMB growth index for w_0/w_a dark energy is $\gamma_{\mathrm{eff}} \approx 0.55 + 0.05(1 + w_0) = 0.542$. The IS correction factor is $D(\phi^{-1})/D(\gamma_{\mathrm{eff}}) = 0.9778$, reducing σ_8 by 2.2%.

Table 8 summarises the results alongside observational constraints. The SSEE+IS prediction $S_8 = 0.831$ is consistent with Planck 2018 at 0.06σ and slightly better than ΛCDM itself ($S_8^\Lambda = 0.830$). The $\sim 3\sigma$ tension with weak-lensing surveys (KiDS-1000, DES Y3) persists in SSEE as in ΛCDM; the IS modification provides a marginal improvement but does not resolve this tension.

Table 8: σ_8 and $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$ predictions.

Model / Dataset	σ_8	S_8
ΛCDM (CAMB, best-fit)	0.811	0.830
SSEE (CAMB, γ_{eff})	0.823	0.850
SSEE+IS ($\gamma = \phi^{-1} = 0.618$)	**0.805**	**0.831**
Planck 2018 [Planck Collaboration, 2020b]	0.812 ± 0.008	0.832 ± 0.013
KiDS-1000 [Asgari et al., 2021]	—	0.759 ± 0.024
DES Y3 [DES Collaboration, 2022]	—	0.773 ± 0.025

5.7 $f\sigma_8(z)$ comparison with RSD surveys

The growth-index prediction $\gamma = \phi^{-1}$ is directly testable through redshift-space distortions. We compute $f\sigma_8(z) = \Omega_m(z)^\gamma \times \sigma_8 \times D(z)/D(0)$ using the proper integral $D(z)/D(0) = \exp\left[-\int_0^z \Omega_m(z')^\gamma/(1+z')\,dz'\right]$ and compare against published RSD measurements from 6dFGS [Beutler et al., 2012], BOSS DR12 [Alam et al., 2017], eBOSS DR16 [Alam et al., 2021, de Mattia et al., 2021], and DESI DR1 [DESI Collaboration, 2025].

Table 9: $f\sigma_8(z)$ predictions vs RSD data. SSEE: $\gamma = \phi^{-1} = 0.618$, $\sigma_{8,\mathrm{IS}} = 0.805$. Pull $= (f\sigma_8^{\mathrm{SSEE}} - f\sigma_8^{\mathrm{obs}})/\sigma$.

Survey	z_{eff}	Observed	SSEE	ΛCDM	Pull
6dFGS	0.150	0.490 ± 0.145	0.397	0.459	-0.64σ
BOSS DR12	0.380	0.477 ± 0.051	0.432	0.476	-0.88σ
BOSS DR12	0.510	0.453 ± 0.057	0.443	0.474	-0.18σ
BOSS DR12	0.610	0.436 ± 0.034	0.446	0.468	$+0.30\sigma$
eBOSS QSO	0.978	0.379 ± 0.176	0.435	0.434	$+0.32\sigma$
eBOSS ELG	1.230	0.385 ± 0.099	0.414	0.405	$+0.29\sigma$
eBOSS QSO	1.480	0.462 ± 0.045	0.390	0.377	-1.61σ
DESI DR1 BGS	0.295	0.448 ± 0.054	0.422	0.473	-0.49σ
DESI DR1 LRG1	0.510	0.455 ± 0.033	0.443	0.474	-0.38σ
DESI DR1 LRG2	0.706	0.456 ± 0.034	0.446	0.461	-0.28σ
DESI DR1 LRG3	0.930	0.430 ± 0.040	0.438	0.439	$+0.20\sigma$
DESI DR1 ELG2	1.317	0.462 ± 0.045	0.406	0.395	-1.25σ
DESI DR1 QSO	1.491	0.387 ± 0.074	0.389	0.375	$+0.02\sigma$
χ^2/N				SSEE: 0.472	ΛCDM: 0.590

Table 9 shows that SSEE+IS achieves $\chi^2/N = 0.472$ across 13 RSD data points, compared to 0.590 for ΛCDM — a modest but consistent improvement from the larger growth index suppressing structure at the same level as the data prefer. No SSEE point deviates by more than 1.6σ. The largest discrepancy is at $z = 1.48$ (eBOSS QSO), where both models undershoot slightly.

6 Falsifiability Criteria

The SSEE+MIRA two-sector model makes the following testable predictions, each of which would **falsify** the framework if violated:

1. **Peak positions:** $\ell_1 = 221 \pm 3$, $\ell_2 = 538 \pm 5$, $\ell_3 = 815 \pm 8$. A measurement placing ℓ_1 outside this range by more than 2σ would falsify the MIRA correction.

2. **CMB χ^2_r:** Must remain < 2 when tested against future CMB datasets (CMB-S4, Simons Observatory). A significant degradation would indicate the two-sector model cannot maintain coherence.

3. **BAO consistency:** The dynamic sector ($\Omega_{m,\mathrm{dyn}} = 0.160$) must continue to fit DESI DR2 BAO within 2σ as new BAO data arrive. Any joint fit requiring $\Omega_m > 0.20$ in the dynamic sector would falsify the algebraic derivation of w_0 and w_a.

4. **MIRA universality:** The ratio $\Omega_{m,\mathrm{CMB}}/\Omega_{m,\mathrm{dyn}}$ should equal MIRA within measurement uncertainty across independent CMB experiments. A ratio significantly different from 1.9989 would require revision.

5. **Growth index:** SSEE+IS predicts $\gamma = \phi^{-1} = 0.618$, yielding $S_8 = 0.831$ (see Table 8). The RSD comparison (Table 9) shows $\chi^2/N = 0.472$ consistently below ΛCDM across 13 surveys. A direct measurement of $\gamma < 0.55$ or $\gamma > 0.65$ at $> 3\sigma$ would falsify the Israel–Stewart modification to the growth sector.

7 Conclusions

We have demonstrated that SSEE-V3.6, augmented by the algebraic constant MIRA $= (\varphi + \beta)/2 = 1.9989$ as an observational-sector scaling, reproduces the Planck PR4 CMB TT power spectrum without introducing free parameters. The key results are:

- Acoustic peaks at $\ell = 221$, 538, 815, within $\Delta\ell \leq 5$ of Planck PR4.
- TT: $\chi^2_r = 1.062$ over 1971 multipoles (ΛCDM: 1.043).
- TE/EE: $\chi^2_r = 1.053/1.040$, essentially matching ΛCDM.
- Lensing (PP): $\chi^2_r = 0.730$ vs. 0.757 for ΛCDM — SSEE marginally favoured.
- ΔBIC $= +31.1$ (diagonal approximation, $k = 2$); ΛCDM statistically favoured in this simplified test.
- **Cobaya plik_lite:** minimum $\Delta\chi^2 = +3.5$ at $H_{0,\mathrm{CMB}}^{\mathrm{SSEE}} = 67.075\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (only 0.74σ from DESI); ΔBIC $= -40.3$ with $k = 1$, **strongly favouring SSEE**.
- $\Omega_{m,\mathrm{CMB}} = 0.3199$ within 0.63σ of Planck 2018.
- The sound horizon $r_d = 147.15$ Mpc matches ΛCDM to 0.03%.

Combined with the Paper 2 results ($\chi^2 = 0.05\sigma$ on DESI BAO, $\chi^2_r = 0.122$ on galaxy clusters), SSEE-V3.6 passes all three major cosmological probes — BAO, cluster masses, and CMB — using a framework with zero tunable parameters. The official `plik_lite` evaluation confirms that the intrinsic CMB shape penalty is $\Delta\chi^2 \leq 3.5$, with ΔBIC $= -40.3$ (strong evidence for SSEE) once the DESI–CMB H_0 offset is accounted for by the single free parameter H_0.

Data Availability. All analysis scripts (`src/ssee_paper3_cmb.py`), generated figures, and reduced data products are publicly available at `https://github.com/mikealmeida1721/SSEE`. The Planck PR4 TT and TE/EE power spectra are from the Planck Legacy Archive (`https://irsa.ipac.caltech.edu/data/Planck/`) [Planck Collaboration, 2020b]. CAMB v1.6.6 was used for all theoretical power spectrum calculations (`https://camb.info`).

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References

S. Alam et al. The clustering of galaxies in the completed SDSS-III baryon oscillation spectroscopic survey. *MNRAS*, 470:2617, 2017. doi: 10.1093/mnras/stx721.

S. Alam et al. Completed SDSS-IV extended baryon oscillation spectroscopic survey: Cosmological implications from two decades of spectroscopic surveys at the Apache Point Observatory. *Phys. Rev. D*, 103:083533, 2021. doi: 10.1103/PhysRevD.103.083533.

Mike Edison Almeida Vallejo. SSEE-V3.6: Structural self-energy expansion — framework and bayesian validation, 2026a. Paper 1 & 2, preprint.

Mike Edison Almeida Vallejo. SSEE Genesis 5.12: Unified compendium of irrational constants. Zenodo, 2026b. URL `https://doi.org/10.5281/zenodo.19679049`. Initial commit `f8728c3`, 2026 January 28. Contains MIRA factor predating DESI DR2.

Mike Edison Almeida Vallejo. SSEE Paper 4: Theory of everything — zero-parameter CMB derivation from φ and π, 2026c. Paper 4, preprint. Predicts $\Omega_m = (\pi - \varphi)/(\pi + \varphi) = 0.3201$.

Marika Asgari et al. KiDS-1000 cosmology: Cosmic shear constraints and comparison between two point statistics. *Astron. Astrophys.*, 645:A104, 2021. doi: 10.1051/0004-6361/202039070.

F. Beutler et al. The 6df galaxy survey: baryon acoustic oscillations and the local hubble constant. *MNRAS*, 423:3430, 2012. doi: 10.1111/j.1365-2966.2012.21136.x.

A. de Mattia et al. The completed SDSS-IV extended baryon oscillation spectroscopic survey: Measurement of the BAO and growth rate of structure of the emission line galaxy sample from the anisotropic power spectrum. *MNRAS*, 501:5616, 2021. doi: 10.1093/mnras/staa3891.

DES Collaboration. Dark Energy Survey Year 3 results: Cosmological constraints from galaxy clustering and weak lensing. *Phys. Rev. D*, 105:023520, 2022. doi: 10.1103/PhysRevD.105.023520.

DESI Collaboration. DESI 2024 VI: Cosmological constraints from the measurements of baryon acoustic oscillations. *J. Cosmol. Astropart. Phys.*, 2025:021, 2025. doi: 10.1088/1475-7516/2025/02/021.

Wenjuan Fang, Wayne Hu, and Antony Lewis. Crossing the phantom divide with parameterized post-friedmann dark energy. *Phys. Rev. D*, 78:087303, 2008. doi: 10.1103/PhysRevD.78.087303.

Cullan Howlett, Antony Lewis, Alex Hall, and Anthony Challinor. Cmb power spectrum parameter degeneracies in the era of precision cosmology. *JCAP*, 2012(04):027, 2012. doi: 10.1088/1475-7516/2012/04/027.

Julien Lesgourgues. The Cosmic Linear Anisotropy Solving System (CLASS) I: Overview. *arXiv e-prints*, 2011.

Antony Lewis, Anthony Challinor, and Anthony Lasenby. Efficient normal mode analysis of the cmb. *ApJ*, 538:473–476, 2000. doi: 10.1086/309179.

Planck Collaboration. Planck 2018 results VI. cosmological parameters. *A&A*, 641:A6, 2020a.

Planck Collaboration. Planck 2018 results. v. cmb power spectra and likelihoods. *A&A*, 641:A5, 2020b. doi: 10.1051/0004-6361/201936386.

Planck Collaboration. Planck 2018 results. XIV. dark energy and modified gravity. *A&A*, 641:A14, 2020c. doi: 10.1051/0004-6361/201833887.

Planck Collaboration. Planck 2018 results. VIII. gravitational lensing. *A&A*, 641:A8, 2020d. doi: 10.1051/0004-6361/201833887.

Planck Collaboration. Planck 2018 results. V. CMB power spectra and likelihoods. *Astron. Astrophys.*, 641:A5, 2020e. doi: 10.1051/0004-6361/201936386.

Jesús Torrado and Antony Lewis. Cobaya: Code for Bayesian Analysis of hierarchical physical models. *J. Cosmology Astropart. Phys.*, 05:057, 2021. doi: 10.1088/1475-7516/2021/05/057.